PythonPing Network Scanner Multithreaded

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Time required: 60 minutes

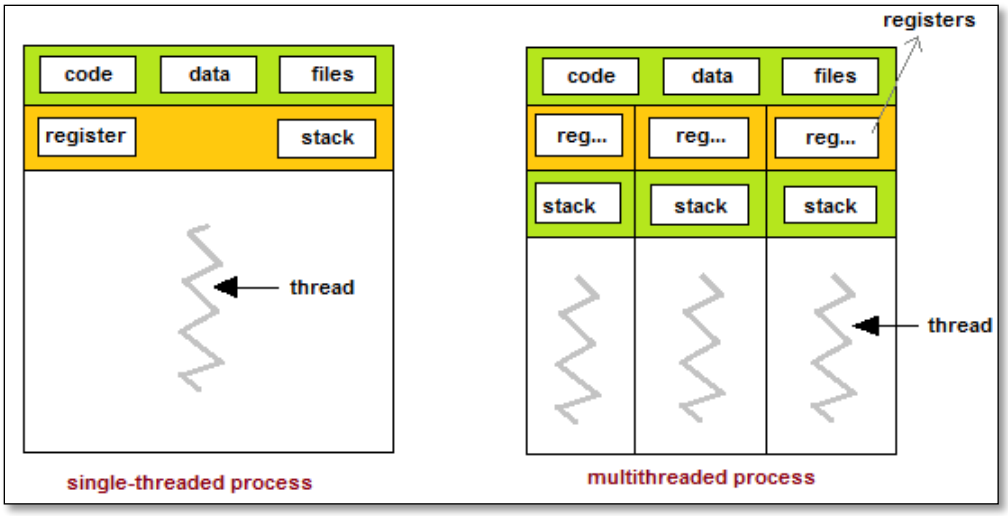
Objective: Write a cross platform Python script that uses branching, looping, multithreading, and pythonping to scan a local network.

Python Threading Tutorial

- https://www.pythontutorial.net/python-concurrency/python-threadpoolexecutor/

Threading

Threading in python is used to run multiple threads (tasks, function calls) at the same time. Python threads are used in cases where the execution of a task involves some waiting. One example would be interaction with a service hosted on another computer, such as a webserver. Threading allows python to execute other code while waiting.



Tutorial 1: Threads, Locks and Queues

```python
#!/usr/bin/env python3
"""
    Filename: threading_example_4.py
"""

import threading        # https://docs.python.org/3/library/threading.html
import queue            # https://docs.python.org/3/library/queue.html
import time             # https://docs.python.org/3/library/time.html


class ThreadExample():
    def __init__(self):
        # Define a thread lock to prevent threads running into each other
        self.thread_lock = threading.Lock()
        # Create thread queue to keep track of the threads
        self.q = queue.Queue()
        # Define number of threads
        number_of_threads = 5
        # Create/spawn multiple threads
        for r in range(number_of_threads):
            # Set the thread target
            thread = threading.Thread(target=self.worker)
            # All threads will end when main program ends
            # Creates a cleaner shutdown
            thread.daemon = True
            # Start the thread
            thread.start()
```

threading.Thread(target=worker) sets the target method for the threads

thread.daemon = True creates a cleaner shutdown. All threads end when the main program ends.

thread.start() spawns the specified number of threads. These threads take turns going through the worker queue.

```
        # Start timer before sending tasks to the queue
        start_time = time.time()

        print(f"Creating a task request for each item in the given range\n")

        # Put all task requests into the queue
        for item in range(10):
            self.q.put(item)

        # Block until all worker tasks are complete in the queue
        self.q.join()
        # Calculate elapsed time for process
        elapsed_time = round(time.time() - start_time, 2)
        print(
            f"All workers completed their tasks after {elapsed_time} seconds")
```

q.put() puts all items into the queue.

q.join() waits until the queue is empty before performing other operations.

When you call **q.join()** in the main thread, it block's the main threads until the workers have processed everything that's in the queue. It does not stop the worker threads, which continue executing their infinite loops. Daemon automatically quit when they are done. When all the work threads have joined, the program continues.

```python
    def worker(self):
        """This method does all the work"""
        while True:
            # Get the next task in the queue
            item = self.q.get()
            # thread_lock prevents the threads from running into each other
            with self.thread_lock:
                print(f'Working on {item}')
                print(f'Finished {item}')
            time.sleep(1)
            # Remove task from queue
            self.q.task_done()


thread_example = ThreadExample()
```

q.get() gets the next item in the queue to work on.

thread_lock prevents the threads from running over each other. Without this, the results would be printed on top of each other.

q.task_done() lets worker threads say when a task is done. It deletes an element from the queue. At the end of the join, the queue length is determined based on whether the queue length is zero. After that the main thread is executed.

There are 5 threads. Each worker task takes 1 second. The run time is 2 seconds.

Example run:

```
Creating a task request for each item in the given range

Working on 0
Finished 0
Working on 1
Finished 1
Working on 2
Finished 2
Working on 3
Finished 3
Working on 4
Finished 4
Working on 5
Finished 5
Working on 6
Finished 6
Working on 7
Finished 7
Working on 8
Finished 8
Working on 9
Finished 9
All workers completed their tasks after 2.0 seconds
```

Find Your Network IP Address in Windows

Use the network address of your local network. Example: 192.168.0.0/24

NOTE: 192.168.56.1 is the VirtualBox adapter address, that is not your network address.

1. Enter the following command at the command prompt: **ipconfig /all**
2. The screenshot below shows my network at home, 192.168.9.0/24 Your IP address will probably be different. I have an Ethernet adapter, you may have a wireless adapter.
3. Notice that my IP address information includes a Default Gateway, DHCP Server, and DNS Servers. Those are needed for a functioning network connection.
4. Note that my IPv4 Address for my computer is **192.168.9.101** My subnet Mask is **255.255.255.0** This makes my network a standard Class C network.
    a. My network address is **192.168.9.0/24**

NOTE: If you are not sure about your network address, please contact me. You will get a 0 for this assignment if you do not provide a screenshot showing a successful scan of your network.

```
C:\Users\Bill.THECOMPUTERGUY>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : Bill-PC
   Primary Dns Suffix  . . . . . . . : thecomputerguy.local
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No
   DNS Suffix Search List. . . . . . : thecomputerguy.local
                                       lan

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . : lan
   Description . . . . . . . . . . . : Realtek PCIe GbE Family Controller
   Physical Address. . . . . . . . . : 2C-F0-5D-A2-AC-3E
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::b08b:b38e:4b9d:3e9b%7(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.9.101(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Friday, April 15, 2022 6:32:36 AM
   Lease Expires . . . . . . . . . . : Sunday, April 17, 2022 6:32:37 AM
   Default Gateway . . . . . . . . . : 192.168.9.1
   DHCP Server . . . . . . . . . . . : 192.168.9.1
   DHCPv6 IAID . . . . . . . . . . . : 103608413
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-27-89-4B-A4-2C-F0-5D-A2-AC-3E
   DNS Servers . . . . . . . . . . . : 192.168.9.10
                                       8.8.8.8
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter VirtualBox Host-Only Network:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VirtualBox Host-Only Ethernet Adapter
   Physical Address. . . . . . . . . : 0A-00-27-00-00-0F
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::b0d1:22cf:dacc:d009%15(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.56.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
   DHCPv6 IAID . . . . . . . . . . . : 168427559
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-27-89-4B-A4-2C-F0-5D-A2-AC-3E
   DNS Servers . . . . . . . . . . . : fec0:0:0:ffff::1%1
                                       fec0:0:0:ffff::2%1
                                       fec0:0:0:ffff::3%1
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

Tutorial 2: PythonPing Threaded Network Scanner

Security professionals often need to automate or create tools to help them conduct security tests. In this activity, you write a Python script that uses the ping command, threading, and a for loop to ping IP numbers for an entire class C network.

1. Copy **pythonping_scanner.py**. Name it: **pythonping_scanner_threaded.py**
2. Modify your code as follows including the comments.

```python
#!/usr/bin/python3
"""
    Filename: port_scanner_threaded.py
    This program prompts the user to enter network address
    it uses the pythonping library to detect active devices
    at each possible IP address in the range
"""

import time
import threading
import queue
# https://docs.python.org/3/library/ipaddress.html
# Convert ip/mask to list of hosts
import ipaddress
# pip install pythonping
from pythonping import ping


class PythonPingScanner():
    def __init__(self):
        # Define a thread lock to prevent threads running into each other
        self.thread_lock = threading.Lock()
        # Create thread queue to keep track of the threads
        self.q = queue.Queue()
        # Simultaneous threads, you can increase or decrease this
        self.NUMBER_OF_THREADS = 50
        print("+---------------------------------+")
        print("|     Threaded Network Scanner     |")
        print("+---------------------------------+")
        self.start_scan()
```

```python
# -------------------------- START SCAN ---------------------------------#
    def start_scan(self):
        """Get network address x.x.x.x/x or x.x.x.x/x.x.x.x from user"""
        self.count = 0
        # --------------------- FIND NETWORK ADDRESS --------------------#
        # Use ipconfig in Windows, ifconfig in Linux
        # to find your local network address
        # Example: If your IP address is 192.168.1.1
        # Subnet mask: 255.255.255.0
        # Your network address is 192.168.1.0/24
        # If your subnet mask is different than 255.255.255.0
        # Type in the subnet mask directly: 192.168.10.0/255.255.255.252.0

        # Change this to the default value of your network
        default_local_network = "192.168.9.0/24"
        # Prompt the user to input a network address and press Enter
        # If they press enter without an network address, the default is used
        network_address = input(
            "\n Enter your network address (ex. 192.168.9.0/24): "
        ) or default_local_network

        # Create a network address object from user input
        ip_net = ipaddress.ip_network(network_address)
        print(f" Ping Scan: {network_address}")
        # Convert all hosts on entered network into a list
        self.all_hosts = list(ip_net.hosts())
        # For debugging
        # print(self.all_hosts)
```

```python
        # Store start time of program scan execution
        start_time = time.time()

        # Create/spawn multiple threads
        for r in range(self.NUMBER_OF_THREADS):
            # Set the thread target method
            thread = threading.Thread(target=self.worker)
            # All threads end when main program ends for cleaner shutdown
            thread.daemon = True
            # Start/spawn the thread
            thread.start()

        # Put all task requests into the queue
        for host in self.all_hosts:
            self.q.put(str(host))

        # Block program from continuing
        # until all worker tasks are complete in the queue
        self.q.join()

        # Calculate elapsed time for process
        scan_time = time.time() - start_time

        print(f" {self.count} hosts found.")
        print(f" Time taken: ({round(scan_time, 2)})sec")
```

```python
# --------------------------- THREAD WORKER -------------------------------#
    def worker(self):
        while True:
            # Get the next IP address from the queue
            host = self.q.get()
            # Scan the IP address
            self.scan(host)
            # Worker announces the task is done, task is removed from queue
            self.q.task_done()

# --------------------------- SCAN NETWORK -------------------------------#
    def scan(self, ip):
        """Ping all IP addresses"""
        result = ping(
            ip,         # Target IP address
            count=2,    # Number of pings
            timeout=2   # Timeout in seconds
        )
        # If response time is less than 2000 ms, the host is alive
        if result.rtt_avg_ms < 2000:
            # thread_lock prevents the threads from running into each other
            with self.thread_lock:
                # Track count of live hosts
                self.count += 1
                # Response time less than 2000ms, target is active
                print(f" {ip:14}-> RTT: {result.rtt_avg_ms:>6.2f}ms")

 # Create program object to start program
python_ping_scanner = PythonPingScanner()
while True:
    menu = input(" Another scan (Y/N):").lower()
    if menu == "n":
        break
    python_ping_scanner.start_scan()
```

Example run:

```
+----------------------------+
|  Threaded Network Scanner  |
+----------------------------+

 Enter your network address (ex. 192.168.1.0/24):
 Ping Scan: 192.168.9.0/24
 192.168.9.10  -> RTT:   0.66ms
 192.168.9.1   -> RTT:   0.89ms
 192.168.9.112 -> RTT:   0.07ms
 192.168.9.113 -> RTT:   2.08ms
 192.168.9.103 -> RTT:   7.68ms
 192.168.9.134 -> RTT:   0.79ms
 192.168.9.129 -> RTT:   1.17ms
 192.168.9.137 -> RTT:   1.79ms
 192.168.9.130 -> RTT:  61.50ms
 192.168.9.139 -> RTT:  71.71ms
 192.168.9.136 -> RTT:  99.80ms
 192.168.9.245 -> RTT:   4.33ms
 12 hosts found.
 Time taken: (20.31)sec
 Another scan (Y/N):n
```

Jazzed up version with rich library.

```
  Python Threaded Network Ping Scanner
            By William Loring
Enter Network (192.168.1.0/24):

Ping Scan: 192.168.9.0/24
192.168.9.1   -> RTT:   0.35ms
192.168.9.10  -> RTT:   0.40ms
192.168.9.101 -> RTT: 248.33ms
192.168.9.103 -> RTT:  19.85ms
192.168.9.112 -> RTT:   0.21ms
192.168.9.113 -> RTT:   2.07ms
192.168.9.134 -> RTT:   1.03ms
192.168.9.129 -> RTT:   2.68ms
192.168.9.137 -> RTT:   2.58ms
192.168.9.139 -> RTT:   2.14ms
192.168.9.130 -> RTT:  19.98ms
192.168.9.136 -> RTT:  21.06ms
192.168.9.245 -> RTT:   1.35ms
13 live hosts
Run Time: 40.4 seconds
Another scan (Y/N):
```

Assignment Submission

1. Attach all program files.

2. Attach a screenshot of each successful program run.

3. If you do not attach a screenshot of a successful program run on your correct network address, you will receive a 0 for this assignment.
4. Submit the assignment in Blackboard.